KEMENTERIAN PENDIDIKAN TINGGI, SAINS, DAN TEKNOLOGI
POLITEKNIK NEGERI MALANG
JURUSAN TEKNOLOGI INFORMASI
Jl. Soekarno Hatta No. 9, Jatimulyo, Lowokwaru, Malang 65141
Telp. (0341) 404424 – 404425, Fax (0341) 404420
https://jti.polinema.ac.id

Mata Kuliah	:	Praktikum-Pemrograman Berbasis Objek
Program Studi	:	D4 – Teknik Informatika
Semester	:	3

Kelas	:	TI 2D
NIM	:	254107020077
Nama	:	Naufal Farhan Nur Ramadhan
Jobsheet Ke-	:	01

Laporan Jobsheet

Perbedaan Procedural dan OOP

Langkah	Jawaban/Deskripsi
1.	Procedural

```java
public class Procedural{
    Run | Debug
    public static void main(String[] args) {
        String itemName1 = "Coffer";
        float price1 = 120000.0f;
        int quantity1 = 3;

        String itemName2 = "Sugar";
        float price2 = 30000.0f;
        int quantity2 = 1;

        System.out.println(itemName1 + " : " + total(price1,quantity1) );
        System.out.println(itemName2 + " : " + total(price2,quantity2) );
    }
    static float total(float price , int quantity){
        return price * quantity;
    }
}
```

2	OOP
	`class Product{`
	`    private final String itemName;`
	`    private final float price;`
	`    static int totBrg=0;`
	
	`    public Product(String itemName, float price ){`
	`        this.itemName = itemName;`
	`        this.price = price;`
	`        totBrg++;`
	`    }`
	`    public float total(int quantity){`
	`        return (quantity* price);`
	`    }`
	`    public String getName(){`
	`        return itemName;`
	`    }`
	`    public static void totBelanjaan(){`
	`        System.out.println("Total Belanjaan : " + totBrg);`
	`    }`
	`}`
	`public class OOP{`
	`    public static void main(String[] args) {`
	`        Product product1 = new Product("Coffe",120000.0f);`
	`        Product product2 = new Product("Sugar",30000.0f);`
	`        Product product3 = new Product("Beras",80000.0f);`
	
	`        System.out.println("Barang ke-1 = "+product1.getName()+" Dengan Total : "+product1.total(3));`
	`        System.out.println("Barang ke-2 = "+product2.getName()+" Dengan Total : "+product1.total(1));`
	`        System.out.println("Barang ke-3 = "+product3.getName()+" Dengan Total : "+product1.total(5));`
	
	`        Product.totBelanjaan();`
	`    }`

Praktikum Percobaan 1

Langkah	Jawaban/Deskripsi

1.	Bike
	`public class Bike {`
	`    private String brand;`
	`    private int speed;`
	`    private int gear=1;`
	`    // Gear 1 : max 5km/h, Gear 2 : max 10km/h, .. Gear 6 : max 60km/h`
	`    private final int[] GEAR_SPEED_LIMITS = {5,10,25,30,40,60};`
	
	`    public void setBrand(String brandName){`
	`        this.brand = brandName;`
	`    }`
	`    public void gearChanges(int gearValue){`
	`        if(gearValue < 1 || gearValue > 6){`
	`            System.out.println("Invalid Gear Value, Gear must between 6 and 1");`
	`        } else {`
	`            gear = gearValue;`
	`        }`
	`    }`
	`    public int speedAcceleration(int increment){`
	`        speed += increment;`
	`        if(speed > GEAR_SPEED_LIMITS[gear-1]){`
	`            speed = GEAR_SPEED_LIMITS[gear-1];`
	`        }`
	`        return speed;`
	`    }`
	`    public int speedDeceleration(int decrement){`
	`        speed -= decrement;`
	`        if(speed < 0){`
	`            speed = 0;`
	`        }`
	`        return speed;`
	`    }`

```
    public void printInfo(){
        System.out.println("Brand : "+ brand);
        System.out.println("Speed : "+speed);
        System.out.println("Gear : "+gear   );
    }
```

2 BikeDemo

```
public class BikeDemo {
    public static void main(String[] args) {
        Bike mountainBike1 = new Bike();
        Bike mountainBike2 = new Bike();
        RoadBike roadBike1 = new RoadBike();

        mountainBike1.setBrand("Trek");
        mountainBike1.speedAcceleration(10);
        mountainBike1.gearChanges(2);
        mountainBike1.printInfo();

        mountainBike2.setBrand("Giant");
        mountainBike2.speedAcceleration(20);
        mountainBike2.gearChanges(7);
        mountainBike2.printInfo();

        roadBike1.setBrand("Specialized");
        roadBike1.setTireWidth(25);
        roadBike1.speedAcceleration(15);
        roadBike1.gearChanges(4);
        roadBike1.printInfo();
    }
}
```

3 Hasil :

```
Brand : Trek
Speed : 5
Gear : 2
Invalid Gear Value, Gear must between 6 and 1
Brand : Giant
Speed : 5
Gear : 1
```

Praktikum Percobaan 2

Langkah	Jawaban/Deskripsi
1.	RoadBike

```
public class RoadBike extends Bike {
    private int tireWidth;

    public void setTireWidth(int width){
        tireWidth = width;
    }

    @Override
    public void printInfo(){
        super.printInfo();
        System.out.println("Tire Width : "+tireWidth+" mm");
        System.out.println(x: "Bike Type : Road Bike");
    }
}
```

2	Hasil :
	`Brand : Trek`
	`Speed : 5`
	`Gear : 2`
	`Invalid Gear Value, Gear must between 6 and 1`
	`Brand : Giant`
	`Speed : 5`
	`Gear : 1`
	`Brand : Specialized`
	`Speed : 5`
	`Gear : 4`
	`Tire Width : 25 mm`
	`Bike Type : Road Bike`

Tugas Praktikum

Langkah	Pertanyaan/Perintah	Jawaban
1.	Foto 4 buah objek di sekitar kalian dengan 2 objek di antaranya merupakan objek yang mengandung konsep pewarisan (inheritance), contoh: kulkas, kursi, meja ruang tamu, meja belajar sehingga diketahui meja ruang tamu dan meja belajar mewarisi objek meja!	    1.Kacamata, 2.Mouse, 3.KamataKamera(turunan), 4.MouseGaming(turunan)
2	Lakukan pengamatan terhadap 4 objek tersebut untuk menentukan atribut dan methodnya!	1.Kacamata -Atribut{warna, mines, tipeLensa, diameter} -Method{diPakai, gantiLensa, diLepas} 2.Mouse -Atribut{Sensi, Warna, Merk, Status} -Method{turnOn, turnOff, adjust,}

		3.KacamataKamera(turunan) -Atribut{ warna, mines, tipeLensa, diameter, baterai,} - Method{diPakai, diBersihkan, gantiLensa, diLepas, rekam, charge} 4. MouseGaming(turunan) -Atribut{Sensi, Warna, Merk, Status, lampuRGB, macro} -Method{turnOn, turnOff, adjust, autoClick, gantiLampu}

3	Berdasarkan 4 buah objek tersebut, buat class nya dalam Bahasa pemrograman Java!	`public class Kacamata {`
		`public class Mouse {`
		`public class MouseGaming extends Mouse {`
		`public class KacamataKamera extends Kacamata{`

4	Perlu diperhatikan bahwa terdapat dua class hasil pewarisan sehingga perlu menambah satu class baru sebagai class yang mewarisi dua class tersebut!	

5	Tambahkan dua atribut untuk setiap class!	`public class Kacamata {`
		`String warna;`
		`int mines;`
		`String tipeLensa = "Blue-ray";`
		`String status = "off";`
		`public class Mouse {`
		`private int sensi = 50;`
		`private String merk;`
		`private String warna;`
		`private String status = "off";`
		`public class KacamataKamera extends Kacamata{`
		`private String merkKamera;`
		`private int resolusi;`
		`public class MouseGaming extends Mouse {`
		`private String lampuRGB;`
		`private int macro;`

6	Tambahkan tiga method untuk setiap class termasuk method cetak informasi!	

```java
public class Kacamata {
    String warna;
    int mines;
    String tipeLensa = "Blue-ray";
    String status = "off";

    public Kacamata(String warna, int mines){
        this.warna = warna;
        this.mines = mines;
    }
public void diLepas(){
    if(status.equalsIgnoreCase(anotherString: "on")){
            status = "off";
        }else{
            System.out.println(x: "KacaMata sudah Terlepas!!..");
        }
}
public void diPakai(){
    if(status.equalsIgnoreCase(anotherString: "off")){
            status = "on";
        }else{
            System.out.println(x: "KacaMata sudah Terpakai!!..");
        }
}
public void gantiLensa(int newMines, String newLensa){
    this.tipeLensa = newLensa;
    this.mines = newMines;
}
public void getInfo(){
    System.out.println(x: "Kacamata================");
    System.out.println("Warna       :"+warna);
    System.out.println("Mines       : "+mines);
    System.out.println("Tipe Lensa  :"+tipeLensa);
```

```java
public class Mouse {
    private int sensi = 50;
    private String merk;
    private String warna;
    private String status = "off";

    public Mouse(String merk, String color) {
        warna = color;
        this.merk = merk;
    }
    public void adjust(int sensiBaru) {
        sensi = sensiBaru;
    }
    public void turnOn() {
        if (status.equalsIgnoreCase(anotherString: "off")) {
            status = "on";
        } else {
            System.out.println(x: "Mouse Sudah Nyala!!..");
        }
    }
    public void turnOff() {
        if (status.equalsIgnoreCase(anotherString: "on")) {
            status = "off";
        } else {
            System.out.println(x: "Mouse Sudah Mati!!..");
        }
    }
    public void getInfo() {
        System.out.println("Merk Mouse      : " + merk);
        System.out.println("Status          : " + status);
        System.out.println("Sensi Mouse     : " + sensi);
        System.out.println("Warna Mouse     : " + warna);
    }
```

```java
public class KacamataKamera extends Kacamata{
    private String merkKamera;
    private int resolusi;

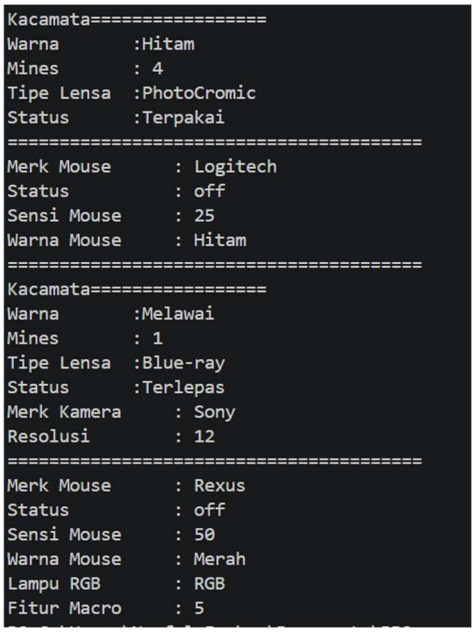
    public KacamataKamera(String merk, int mines, String merkKamera, int resolusi) {
        super(merk, mines);
        this.merkKamera = merkKamera;
        this.resolusi = resolusi;
    }
    public void getInfoKamera() {
        super.getInfo();
        System.out.println("Merk Kamera     : " + merkKamera);
        System.out.println("Resolusi        : " + resolusi);
    }
}
```

```java
public class MouseGaming extends Mouse {
    private String lampuRGB;
    private int macro;

    public MouseGaming(String merk, String warna, String lampuRGB, int macro)
        super(merk, warna);
        this.lampuRGB = lampuRGB;
        this.macro = macro;
    }
    public void getInfoGaming() {
        super.getInfo();
        System.out.println("Lampu RGB       : " + lampuRGB);
        System.out.println("Fitur Macro     : " + macro);
    }
}
```

String lampuRGB

Source: PBO_550435e7

7	Tambahkan satu class Demo sebagai main!	 

```
public class Demo {
    Run | Debug
    public static void main(String[] args) {
        Kacamata km = new Kacamata(warna: "Hitam", mines: 2);
        Mouse m     = new Mouse(merk: "Logitech", color: "Hitam");

        KacamataKamera kmk = new KacamataKamera(merk: "Melawai", mines: 1, merkKamera: "Sony", resolusi: 12);
        MouseGaming mg = new MouseGaming(merk: "Rexus", warna: "Merah", lampuRGB: "RGB", macro: 5);

        km.gantiLensa(newMines: 4, newLensa: "PhotoCromic");
        km.diPakai();
        km.getInfo();
        System.out.println(x: "=======================================");
        m.adjust(sensiBaru: 25);
        m.getInfo();
        System.out.println(x: "=======================================");
        kmk.getInfoKamera();
        System.out.println(x: "=======================================");
        mg.getInfoGaming();
    }
}
```

```
Kacamata==================
Warna        :Hitam
Mines        : 4
Tipe Lensa   :PhotoCromic
Status       :Terpakai
=======================================
Merk Mouse       : Logitech
Status           : off
Sensi Mouse      : 25
Warna Mouse      : Hitam
=======================================
Kacamata==================
Warna        :Melawai
Mines        : 1
Tipe Lensa   :Blue-ray
Status       :Terlepas
Merk Kamera      : Sony
Resolusi         : 12
=======================================
Merk Mouse       : Rexus
Status           : off
Sensi Mouse      : 50
Warna Mouse      : Merah
Lampu RGB        : RGB
Fitur Macro      : 5
```

5. Pertanyaan

1. Jelaskan perbedaan antara object dengan class!

Class adalah blueprint nya atau Wadahnya , dan Object adalah Benda nyata nya

2. Jelaskan alasan gear dan brand dapat menjadi atribut dari object Bike!

Karna pada dunia nyata Sepeda pada umumnya memang memiliki Merek dan Gear

3. Sebutkan salah satu kelebihan utama dari pemrograman berorientasi objek dibandingkan dengan pemrograman prosedural!

Kelebihan utama yang dimiliki pemrograman berorientasim objek dibandingkan dengan prosedural adalah ketika kita memerlukan objek atau variable baru kita tidak perlu menambahkan ke kodenya variabel baru, kita cukup membuat objeknya dengan menginisiasinya sekaligus variabel atribut-atributnya.

4. Apakah diperbolehkan melakukan pendefinisian dua buah atribut dalam satu baris kode seperti “public String nama, alamat;”?

Melakukan pendefinisian dua buah atribut dalam satu baris kode diperbolehkan

5. Pada class RoadBike, jelaskan alasan atribut brand, speed, dan gear tidak lagi ditulis di dalam class tersebut!

Pada class Roadbike kita memakai metode pewarisan atau inheritance dengan menuliskan kode extends Bike, yang artinya kita melanjutkan atau mewarisi atribut dan fungsi yang dimiliki oleh class Bike